

Solar Energy for Your Home

You too can harness the power of sun to run your home appliances. Here we share the options to convert solar radiation into electricity, calculation of the solar power needs of your home and some innovative ideas



P. RAJESH, N. PAVAN KUMAR AND SYED ZAMEERUDDIN

There are two major ways to convert the solar radiation into electricity: photovoltaic cells and concentrated solar power. Photovoltaic cells convert the solar power directly into electricity. You may already be familiar with photovoltaic cells if you have ever used a solar calculator. Photovoltaic cells in calculators are a good idea to get rid of batteries but these can't be used for higher power needs owing to their high cost. Typically, the power efficiency of photovoltaic cells is 12-24 per cent, which is lower than other energy sources. The high cost of photovoltaic cells prevents their wide usage but they do find application in areas like satellites, which need continuous power without bat-

teries, and small devices (calculators and watches).

Concentrated solar power (CSP) systems concentrate the sun beams to produce heat and then convert the heat energy into electricity by conventional ways. There are several designs for concentrating the sun beams but the method is the same in all designs: There is a reflector which concentrates the beams in a heat collector, also called absorber. A fluid (molten salt, for example) flowing in the receiver stores the heat. The hot fluid flows to the heat engine and the heat energy is converted into electricity.

CSP system designs

Parabolic-trough design. In parabolic design, there are parabolic mirrors collecting the heat on a receiver that is placed on the mirror's focus. The

energy production is as explained above: The hot fluid is flown through pipes and transferred to the heat engine. This is the most practised method of solar such power plants and there are several such power plants fully operational. Parabolic-trough power plants have an efficiency of 20 per cent, but less area usage and cheap design are their key advantages over other systems.

Power tower design. The power tower design is similar to the parabolic design but it uses a large number of flat, mobile mirrors (also called heliostats) that focus the beams on a single receiver (tower). All the energy focused on a single point creates higher temperatures, thus creating electricity with higher efficiencies. There are several such power tower plants in production. Although parabolic-trough design is proven, the higher efficiencies and lower energy cost predictions are creating more investments for power tower.

Dish design. Dish designs use the same principle as the satellite receivers. The dish collects and focuses the energy at a receiver facing the dish. There are several advantages of dish design: These provide the highest solar-to-electric efficiency (31.25 per cent) among all CSP technologies. Also, each unit has its own standalone structure, which makes it fit for smaller applications (unlike tower design that has a high central tower cost). However, the dish design is costlier than other systems because of its compiler design. Also, it requires more maintenance. Nevertheless, high efficiency is promising a future for dish designs also.